Compute Predictor Collinearity

Description

Analyses collinearity among abiotic predictors in a long-format data frame and returns a 'collinear_output' object produced by 'collinear::collinear()'. The function pivots 'data_source' from long to wide format (one column per variable), removes any 'age' column, screens out zero-variance columns, and then performs the collinearity analysis. Predictors with fewer than two finite observations are removed before their variance is evaluated.

Usage

```
compute_predictor_collinearity(data_source, verbose = TRUE)
```

Arguments

`data_source`	A data frame in long format containing at minimum the columns 'abiotic_variable_name' (character, predictor names) and 'abiotic_value' (numeric, predictor values). An optional 'age' column is silently dropped before analysis.
`verbose`	Logical scalar. If 'TRUE', warn when zero-variance predictors are removed before the collinearity analysis.

Details

The function validates inputs with 'assertthat' and performs post-hoc assertions on the output to guarantee structural integrity before returning. Missing values are filled with 'NA' when pivoting to wide format. The 'age' column is excluded because it is a sampling dimension rather than a predictor. Predictor names are captured before pivoting so that the zero-variance check is scoped to predictor columns only — ID or metadata columns that survive the pivot are never passed to 'collinear::collinear()'. Predictors with insufficient finite observations or zero variance are dropped and reported via 'cli::cli_warn()'. If fewer than three rows are available, or no predictor with non-zero variance remains, the function aborts via 'cli::cli_abort()'.

Value

A 'collinear_output' object as returned by 'collinear::collinear()'. The object contains a 'result' element with a 'selection' character vector of the non-collinear predictor names that were retained.

See Also

[collinear::collinear()] for the underlying collinearity method, [extract_abiotic_data()] for producing the expected input format.